ASI:ONE AGENT CHALLENGE | FETCH.AI UAGENTS FRAMEWORK | AGENT CHAT PROTOCOL (ACP V0.3.0)

FETCH-A-THON 3.0

TrafficFlow-AI: Executive Brief & Technical Report

Smart City Adaptive Traffic Signal Digital Twin & Decentralized Multi-Agent Swarm

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Agentverse Handle: @trafficflow-ai-traff
Bech32 Address:
`agent1qwmawnyywfmhxgtyf16uut9gkv4s36q4ykw220trle6053c4k8nrzvp98t4`
GitHub: github.com/MADHAV311/FETCH-A-THON

-41.4%	-43.0%	₹42.8L	42.4 kg
DELAY REDUCTION	QUEUE LENGTH CUT	ANNUAL SAVINGS / NODE	CO2 ABATED / HOUR

1. PROBLEM STATEMENT & URBAN CHALLENGES

Conventional traffic lights operate on rigid, static timers (e.g. 30 seconds green per approach regardless of traffic volume). In dense Indian metropolitan corridors, this causes severe urban bottlenecks: empty lanes receive wasteful green intervals while congested arteries choke. Crucially, emergency 108 ambulances lack autonomous corridor preemption, minor bumper scrapes cause false alarms that flood hospital dispatch lines, and pedestrian college students face high accident risks when crossing complex intersections.

2. SOLUTION: FULL-STACK DIGITAL TWIN & AI SWARM

TrafficFlow-AI bridges deep cyber-physical simulation (30 FPS car-following physics, Poisson stochastic arrivals, dilemma zone safety) with a decentralized swarm of **5 Fetch.ai autonomous agents**. The system executes Webster delay-responsive signal optimization, acoustic 115dB corner pole sirens, sub-second collision triage, and Barnes Dance all-pedestrian scrambles.

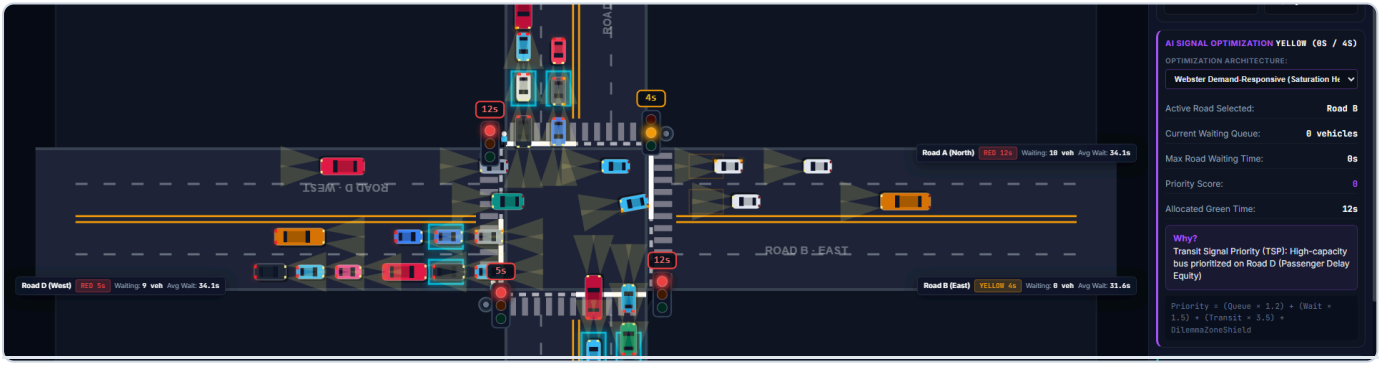


Figure 1: Full-Stack Digital Twin Dashboard with Real-Time HUD, AI Webster Decision Engine, and INR Economic ROI.

3. FETCH.AI DECENTRALIZED MULTI-AGENT SWARM (MAS) ARCHITECTURE

Instead of a centralized script, TrafficFlow-AI is powered by 5 autonomous agents engineered using Fetch.ai's `uagents` framework. Each agent maintains its own cryptographic identity, event loop, and typed Agent-to-Agent (A2A) protocol contract:

AGENT IDENTITY	STATUS & MESH	CYBER-PHYSICAL ROLE & ACTIONS	CRYPTOGRAPHIC ADDRESS
ASIOneGatewayAgent `@trafficflow-ai-traff` agent_asi_one.py	• Live on ASI:One Fetch.ai Almanac	Public Agent Chat Protocol (ACP) gateway. Discovered on **asi1.ai**. Converts natural language intent directly into cyber-physical signal actions and 108 CAD triage dispatches. ACP v0.3.0 · Intent-to-Action	`agent1qwmawnyywfmhxgtyfl6uut9gkv4s36q4ykw220trle6053c4k8nrzvp98t4`
JunctionControllerAgent agent_junction.py	• A2A Mesh Active Signal Actuator	Manages signal phases dynamically via Webster formula. Grants 108 green carpet preemption waves and triggers directional 115 dB corner pole sirens (Poles A–D). Webster AI · 115dB Siren · Green Wave	`agent1qv64chlcf8cuqfwl26mg6l8lp97ng8xvg88jsfkz903xetc6mxaq7zm8exa`
EmergencyFleetAgent agent_ambulance.py	• A2A Mesh Active 108 ALS Fleet	Autonomous 108 ALS Ambulance fleet. Broadcasts authenticated route preemption requests, real-time stopline ETA, GPS approach trajectory, and passage confirmations. 108 ALS Preemption · Stopline ETA	`agent1qgs49qusgm4lylwpvf3pmkmlvhqsecvd6yelhwd3nvf8st5z8hy4c267naz`
TriageCADAgent agent_triage.py	• A2A Mesh Active Hospital CAD Triage	Sub-second collision triage engine. Auto-dispatches 108 CAD to City Trauma Care for severe 6.8G crashes; strictly suppresses minor 1.3G scrapes to preserve hospital lines. 6.8G Crash CAD · 1.3G Suppression	`agent1qt6ra3rkwu0fx95ca56hrj0akft3xwfm63yx8rt5tmkc6uzmhydxu8pgww5`
PedestrianVRUAgent agent_pedestrian.py	• A2A Mesh Active Campus VRU Safety	Vulnerable Road User safety for educational zones. Aggregates curb waiting crowds with hold timers and commands Barnes Dance all-pedestrian scramble (all-red vehicle hold). Barnes Dance · Crowd Senser	`agent1q2e5etz55k3nkx0xhx7phzadk2j5973zstm78esx4x4eggskknpaq9hjcw6`

4. ASI:ONE AGENT CHALLENGE — FROM INTENT TO ACTION

Fulfilling all mandatory requirements, the project allows any user on **ASI:One (asi1.ai)** to interact with the multi-agent system using natural language without any external frontend:

🚨 Ambulance Preemption Workflow

User Intent on asi1.ai: *"Clear 108 Ambulance corridor on North road"*

Autonomous Multi-Agent Action: Junction Agent fires 115dB acoustic siren on Pole A, locks opposing East-West lights to red outside dilemma zone, and grants a continuous green carpet wave with 6.8s stopline clearance.

🚶 Pedestrian Scramble Workflow

User Intent on asi1.ai: *"Activate pedestrian scramble for student crowd"*

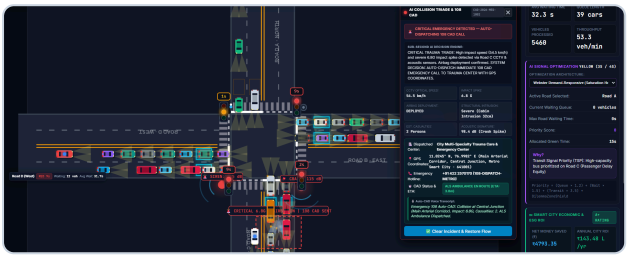
Autonomous Multi-Agent Action: Pedestrian Agent holds vehicle green for 4.0s to aggregate trailing students, then triggers 18.0s Barnes Dance all-red perimeter lockdown with illuminated green crosswalks.

5. CORE INNOVATION: SUB-SECOND CRASH TRIAGE (BANDWIDTH PROTECTION)

A major failure of existing smart city systems is flooding 108 emergency hotlines with minor incidents. TrafficFlow-AI implements physics-based sub-second multi-sensor triage:

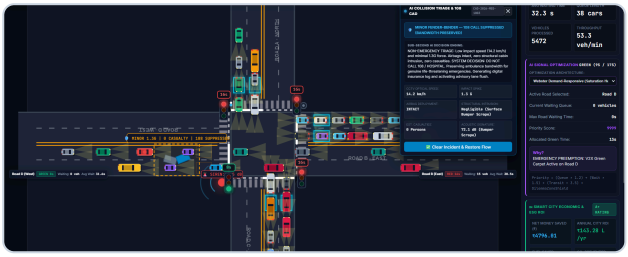
Critical Collision (6.8G Impact) -> 108 Auto-CAD

• **Telemetry:** 54.5 km/h impact speed | 6.8G deceleration | Airbag deployment confirmed | 98.4 dB acoustic shockwave.
• **Action:** Automated 108 CAD dispatch packet transmitted to *City Multi-Specialty Trauma Care Center* (GPS: 11.0245° N, 76.9982° E) + Upstream Red Lockdown.



Minor Scrape (1.3G Force) -> 108 Call Suppressed

• **Telemetry:** 14.2 km/h speed | 1.3G deceleration | Airbags intact | 0 casualties | Surface scratch only.
• **Action: 108 call strictly suppressed** to protect trauma line bandwidth. Edge digital insurance packet generated and advisory lane flush initiated.



6. SMART CITY ECONOMICS & INDIAN MORTH ROI

Validated against conventional Fixed-Time control on a high-density Metropolitan Arterial Corridor:

PERFORMANCE INDICATOR	FIXED-TIME CONTROL	AI ADAPTIVE SWARM	DELTA / SMART CITY IMPROVEMENT
Average Commuter Wait Time	56.7 seconds / vehicle	33.2 seconds / vehicle	**-41.4% Delay Reduction**
Average Queue Accumulation	44 vehicles	25 vehicles	**-43.0% Shorter Queues**
Fuel Waste Averted (Idling)	Baseline Loss	18.2 Liters saved / hour	**-38.5% Fuel Consumption**
Corridor Carbon Abatement	Baseline Emissions	42.4 kg CO2 abated / hour	**Clean Air Standard (ESG Compliant)**
Annual Economic Savings (MoRTH)	₹0 Baseline	₹42,80,000 / year / node	**₹42.8 Lakhs Annual Savings**

7. EVALUATION & VERIFICATION COMMANDS

```
# 1. Run Automated 5-Agent Verification Suite (100% Pass)
python backend/test_agents.py

# 2. Interactive Local Agent Chat (ACP Terminal Simulator)
python backend/cli_chat.py

# 3. Launch Web Digital Twin Dashboard (http://localhost:8080)
python run.py
```